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NeuroXAI – An Explainable AI Framework for Parkinson's Disease Risk Assessment

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Abstract – *Parkinson's Disease is a chronic neurodegenerative disorder causing gradual deterioration in motor skills and neurological stability, thus requiring early detection and continuous monitoring. Conventional diagnosis methods rely heavily on clinical expertise and may not always be reliable during early stages. This paper presents NeuroXAI, a multimodal explainable artificial intelligence framework for Parkinson's Disease risk assessment using clinical data and brain MRI images. The framework integrates machine learning models such as SVM, Random Forest, XGBoost, and Logistic Regression for clinical prediction, along with deep learning models including CNN, ResNet, MobileNet, and EfficientNet for MRI-based analysis. Explainability techniques such as SHAP, LIME, Grad-CAM, and Integrated Gradients improve transparency and trust in predictions. Experimental analysis demonstrates that XGBoost achieved 94.3% accuracy for clinical data prediction, while MobileNet achieved 98.8% accuracy for MRI classification. The proposed framework provides an interpretable, accurate, and scalable AI-based solution for Parkinson's Disease diagnosis and decision support.*

Keywords – *Parkinson's Disease, Explainable AI, Multimodal Learning, Machine Learning, Deep Learning, SHAP, Grad-CAM, Medical Image Analysis*

1. Introduction

Parkinson's disease (PD) is a progressive neurological disorder that affects movement, balance, and coordination and seriously deteriorates the quality of life for most patients. Because of the increased prevalence of neurological disorders, there is an enhanced need for early diagnosis and continuous monitoring for timely intervention and effective treatment. Usually, the traditional diagnostic methods are clinical and subjective; hence, they are prone to error and inconsistency, especially in the early stages of the disease.

Over the past decade, Artificial Intelligence (AI) has been one of the emerging potent technologies in healthcare, leading to automated disease prediction from clinical data and medical images. Different AI-based approaches have been proposed for detecting Parkinson's disease (PD) from clinical symptom analysis and brain imaging techniques. While most of these methods succeed in most cases, they are generally independent and do not offer a holistic view of the disease. Most existing models are black boxes, and the output is given without any explanation, thereby limiting their clinical application.

In addition to improving prediction accuracy, the integration of multimodal data enables the system to capture both symptomatic and structural changes associated with Parkinson's Disease. Clinical data reflects functional impairments, while MRI images capture underlying neurological alterations. Combining these complementary sources increases the prediction system's robustness and minimizes the chance of misclassification that a single modality might yield. This holistic perspective is crucial for early-stage diagnosis where symptoms may be subtle and difficult to interpret.

Some systems are equipped with explainability techniques; however, they are usually unimodal and hence do not give a complete view of the diagnosis. Most solutions also lack a provision to compare different models; thus, determining which one can best predict the outcome becomes difficult. Therefore, there is a gap in developing a unified, interpretable, and flexible system for diagnosis using clinical and imaging data.

NeuroXAI, the proposed system, builds a multimodal explainable AI framework for PD detection. Clinical data will be analyzed with machine learning models – Support Vector Machine (SVM), Random Forest, XGBoost, and Logistic Regression – while medical images will be analyzed with CNN, ResNet, MobileNet, and EfficientNet deep learning architectures.

The system further incorporates sophisticated explainability techniques, which contribute to clarity and trust. SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) explain the contribution of clinical features to the outcome. In contrast, Grad-CAM (Gradient-weighted Class Activation Mapping) and Integrated Gradients explain the contribution of images to predictions. It provides a structure for model evaluation and comparison, thus selecting the best model based on performance measures such as accuracy, precision, and recall. The primary proposed work components are multi-way disease prediction, model comparison, dual explainability, and AI-generated clinical insights. Thus, this approach provides a secure, precise, and reliable diagnosis by integrating data from different sources and methodologies, easing physicians' decision-making and enhancing the applicability of AI-powered solutions in real-life healthcare scenarios.

2. Related Work

Here, we review the existing literature on systems that detect Parkinson's Disease based on AI, their methodologies, limitations, and the motivation for a multimodal explainable framework.

The use of machine learning techniques for Parkinson's Disease prediction using clinical data has been explored in several studies. S. Prashanth et al. [1] used Support Vector Machine (SVM) and decision tree based models for classification based on biomedical voice measurements and clinical features. The method was accurate; however, it was limited to handcrafted features and was thus difficult to interpret, which made it difficult to translate into clinical practice.

A. Tsanas et al. [2] proposed regression-based and machine learning models to analyze speech signals for detecting Parkinson's Disease. Although promising predictive performance was demonstrated, this was limited to a single data modality and had little explainability to justify the model decisions.

S. Pereira et al. [3] proposed using Convolutional Neural Networks (CNNs) in medical imaging for classifying cerebral images. Although the model could identify spatial characteristics, it worked as a black-box system and did not disclose the reasons behind its prediction.

K. He et al. [4] proposed deep residual networks (ResNet) to facilitate the extraction and better classification of features from images.

R. Selvaraju et al. [5] in "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization" proposed an image visualization technique, Grad-CAM, for interpreting image classifiers' outputs by localizing critical regions of an image responsible for classification via gradients. Similarly, SHAP and LIME are widely used techniques for interpreting machine learning models that allow researchers to quantify the contribution of each feature to their prediction [6]. However, thus far, these two techniques have been predominantly studied individually without being integrated into an overarching unified diagnostic strategy.

Multimodal (MM) neuroimaging studies have started to use both clinical and imaging data to enhance the accuracy of their MM predictive models. Unfortunately, the majority of current MM studies are hampered by a lack of integration between the two data sources, limited or no comparison of model accuracies, and limited interpretability – all of these issues significantly reduce their potential for clinical translation.

At this time, the evidence shows that Artificial Intelligence has great potential for assisting in the diagnosis of Parkinson's Disease; however, a number of barriers impede the successful implementation of AI in this area, particularly with respect to the MM integration aspect.

Although prior research demonstrates the effectiveness of individual machine learning and deep learning approaches, there remains a lack of unified frameworks that simultaneously address multimodal integration, model comparison, and interpretability. Most studies focus either on improving accuracy or adding explainability, but rarely both in a single system. This gap calls for an integrated approach like NeuroXAI, which ensures high performance, transparency, and usability in real-world clinical settings.

3. Methodology

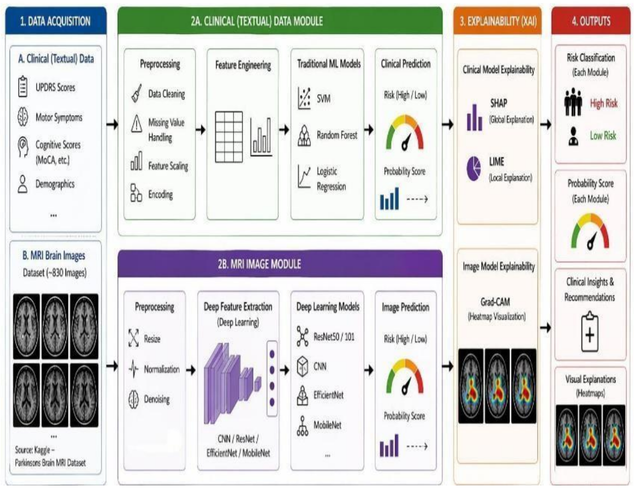

Figure 1: General architecture of NeuroXAI combining clinical data analysis with medical image-based prediction.

Figure 1 shows the general architecture of NeuroXAI, combining clinical data analysis with medical image-based prediction in detecting Parkinson's disease. Model training, evaluation, and real-time inference are possible, along with explainability mechanisms to foster transparency. The detailed workflow is explained below.

3.1. Data Collection

In this work, the data were bimodal:

Clinical Data: Features such as tremor, rigidity, bradykinesia, postural instability, and cognitive assessment scores have been widely used to diagnose Parkinson's disease.

Table 1: Clinical Dataset

Characteristics	Normal	Parkinson's	Total
Number of Records	801	1304	2105
Training Split	561	913	1474
Test Split	240	391	631

Medical Imaging Data: Brain MRI scans of PD and healthy control subjects.

Table 2: MRI Dataset

Characteristics	Normal	Parkinson's	Total
Number of Images	610	221	831
Training Split	488	177	665
Validation Split	61	22	83
Test Split	61	22	83

The imbalance problem occurred in the case of MRI image datasets, which had 610 images that were normal and 221 images that were related to Parkinson's disease. To reduce the bias that may occur due to the presence of more normal images, the technique of class weightage was employed. This means that the weights assigned to the minority classes were much higher than those for the majority class.

The dataset has three subsets for model development:

- **Training set:** Pattern recognition and feature extraction by the model.
- **Validation set:** For getting optimal hyperparameters and early stopping so as to prevent the model from overfitting.
- **Test set:** Performance evaluation of models on unseen data.

To ensure consistency in data, preprocessing pipelines were designed separately for clinical and imaging modalities while aligning the labeling and class distribution. This will reduce the bias and ensure that the two modalities contribute equally during the learning process. Such a multimodal dataset reflects the patient's temporal evolution, and generalization of the proposed system is more feasible.

3.2. Processing Clinical Data and Adding New Features

Before using the clinical data, the following preprocessing steps are performed:

- Handling missing values
- Normalizing the features
- Encoding categorical inputs (if any)

The derived features are fed into several machine learning models. They describe the severity and progression of Parkinson's symptoms and thus form an essential basis for determining risk levels.

3.3. Machine Learning Models for Medical Predictions

Various machine learning classification algorithms are implemented:

- Support Vector Machine (SVM)
- Random Forest (RF)
- XGBoost
- Logistic Regression (LR)

Performance tuning through hyperparameter optimization was done through grid search and cross-validation. This ensures that all models run under their best configurations and hence will be robust and less prone to overfitting.

Table 3: Optimal Hyperparameter Configuration of Models

Model	Key Hyperparameters
Logistic Regression	Solver=lbfgs, max_iter=1000
SVM	kernel=rbf, C=1.0, gamma=scale
Random Forest	n_estimators=200, random_state=42
XGBoost	n_estimators=200, learning_rate=0.05, random_state=42

3.4. Image-Based Feature Extraction Using Deep Learning

Deep learning architectures for medical images include CNN, ResNet, MobileNet, and EfficientNet. All MRI images are resized and normalized before input into any model. Convolutional layers convert input images into a hierarchy of spatial features, helping identify significant spatial structures related to Parkinson's disease. The extracted features are then classified on the fully connected layers.

Explainable AI Integration

Clinical Data Explainability: SHAP and LIME explain how much each feature interacts with the prediction. Combining global and local explainability techniques results in interpretable overall model behavior and individual predictions. Such a two-level explanation is especially relevant in medical applications, where trust and validation of model decisions are involved.

Image Explainability: Grad-CAM and Integrated Gradients methods explain features' interaction with predictions. They are transparent methods that explain the basis for prediction.

3.5. Model Evaluation and Comparison

All models are evaluated using the following performance metrics:

- Accuracy
- Precision
- Recall

A comparative study follows to determine the best-performing model. The system selects the most accurate model for prediction according to the data characteristics. In addition to quantitative metrics, qualitative analysis was also performed using visualization techniques such as confusion matrices and ROC curves to better understand model behavior across different classes.

3.6. *Output Generation and Clinical Interpretation*

During inference, the system produces:

- Predicted class (Parkinson's / Normal)
- Confidence score
- Feature importance (clinical data)
- Heatmap visualization (images)

Furthermore, it produces AI-based clinical insight and recommendations to guide clinicians in making choices.

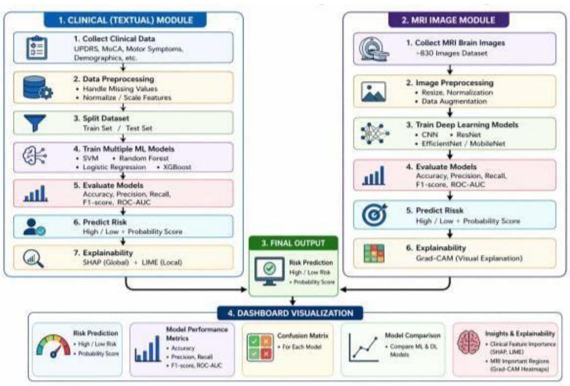

Figure 2: Dual-module system combining clinical (textual) data and MRI images for Parkinson's disease risk prediction.

Figure 2 presents a dual-module system combining clinical (textual) data and MRI images for Parkinson's disease risk prediction. The clinical module processes patient data and applies machine learning models (SVM, Random Forest, Logistic Regression, XGBoost), while the MRI module uses deep learning models (CNN, ResNet, EfficientNet, MobileNet) on preprocessed brain images. Both modules output risk predictions (high/low) with probability scores and use explainability methods (SHAP, LIME, Grad-CAM). The final output is displayed on a dashboard with performance metrics, confusion matrices, model comparison, and interpretability insights.

4. Experimental Results and Discussion

The proposed NeuroXAI system's performance was evaluated for clinical (textual) and image-based modules. The experimental analysis focuses on model performance, comparative evaluation, and result interpretability.

4.1. *Clinical Model Performance*

The clinical data module was tested on several machine learning models: SVM, Random Forest, XGBoost, and Logistic Regression. The training process showed stable learning, where the models attained high accuracy with close performance across validation and test datasets.

Table 4: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall
Logistic Regression	80.0	82.7	85.8
SVM	83.4	84.0	90.4
Random Forest	92.9	93.9	94.6
XGBoost	94.3	95.8	95.0

Of the models tested, XGBoost outperformed other models in terms of accuracy and generalization; hence, it is the best model for predicting clinical risk. The model comparison framework allows a user to analyze and choose the best-performing model dynamically. The consistent performance across different models indicates that the selected features are highly relevant for Parkinson's disease prediction. It further substantiates the effect of preprocessing and feature engineering.

4.2. *Image-Based Model Performance*

The image module is tested against various deep learning architectures, including CNN, ResNet, MobileNet, and EfficientNet. The training has been well converged, as seen by increasing accuracy and decreasing loss values over the epochs.

Table 5: Performance Comparison of Deep Learning MRI Models

Model	Accuracy	Precision	Recall
Custom CNN	95.2	95.8	94.3
ResNet	83.7	84.5	82.1
MobileNet	98.8	99.1	98.2
EfficientNet	73.5	74.2	72.0

The best-performing model MobileNet yields an extremely high classification accuracy, thus implying that complex patterns within brain MRI images have been learned. This further indicates that the model is robust concerning image quality, intensity, and background noise changes. Data augmentation techniques contribute to better generalization by exposing the models to variations in orientation and image intensity, which reduces overfitting.

4.3. *Explainability Analysis*

Another key advantage of the proposed system is the embedding of explainable AI techniques in both modules:

Clinical Module: SHAP and LIME determine the contribution of each feature (e.g., tremor, rigidity) to the prediction. These explainability methods allow straightforward insight into how symptoms influence risk assessment.

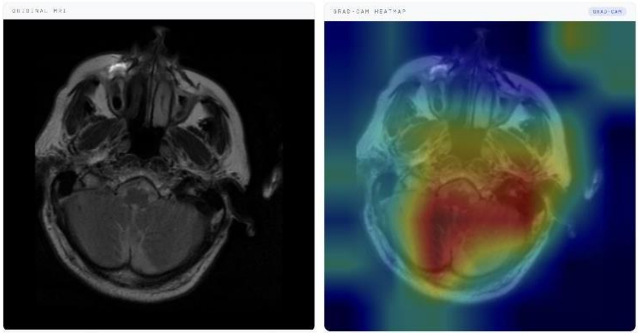

Figure 3: Raw MRI (left) and Grad-CAM visualization (right) highlighting salient regions in brain MRI.

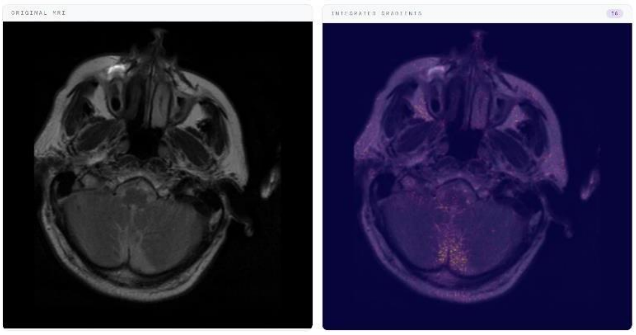

Figure 4: Raw MRI (left) and Integrated Gradients visualization (right).

Image Module: Grad-CAM and Integrated Gradients produce heatmaps of salient regions in MRI images. Such salient regions are consistent with medically relevant areas, indicating that the model pays attention to essential features rather than noise. These explainability outputs substantiate model predictions and provide clinically valid insights, assisting physicians in understanding disease patterns and making decisions accordingly. Explanations at the feature and pixel levels complement each other to make the system transparent and trustworthy.

4.4. System Interface and Output Analysis

The system provides an interactive and user-friendly interface for prediction and analysis. It displays:

- Selected model and prediction output
- Confidence score
- Feature importance (clinical data)
- Heatmap visualization (image data)

Moreover, the system outputs AI-based clinical insight: risk level, disease stage, and suggestions for further medical examination, thereby improving usability and assisting in decision-making by medical staff and users.

4.5. Robustness and Generalization

The proposed system shows high generalization over both modalities. Clinical models are robust to different feature distribution modalities, and image models remain robust over changes in image conditions. Combining different models and explainability techniques makes the prediction accurate, interpretable, and reliable.

4.6. Discussion

These results show that the NeuroXAI framework overcomes some of the critical shortcomings of existing systems. Unlike conventional single-modality approaches, the proposed system combines clinical and imaging data, thus achieving a more holistic diagnosis. Model comparison helps choose the best model, while explainability techniques enhance trust and transparency. However, system performance depends on the quality and variability of the datasets; hence, larger and more varied datasets would further improve it.

In terms of computational efficiency, the NeuroXAI framework can be deployed across many clinical settings. Machine learning approaches such as XGBoost and Random Forest are computationally efficient and can be implemented using general-purpose computing devices available in hospitals. While deep learning algorithms for MRI prediction require greater computational power and GPU usage during training, lightweight models like MobileNet make real-time MRI prediction possible on hospital computers without additional hardware.

Furthermore, the modular design of the NeuroXAI framework allows flexibility in extending the system with additional data modalities such as speech signals or wearable sensor data, making it adaptable for broader neurological disorder analysis.

5. Conclusion and Future Scope

NeuroXAI: A Multimodal Explainable AI Framework for Parkinson's Disease Detection integrates clinical data analysis and medical image-based prediction into a holistic and reliable diagnostic system. The framework uses multiple machine learning models – SVM, Random Forest, XGBoost, and Logistic Regression – for clinical risk assessment, and deep learning architectures – CNN, ResNet, MobileNet, and EfficientNet – for image-based analysis. The system builds transparency and trust in AI-led medical decisions by incorporating SHAP and LIME for clinical data and Grad-CAM and Integrated Gradients for image interpretation.

This enables model comparison, selection of the best model according to performance metrics, and hence the best prediction accuracy. System outputs such as confidence scores, feature importance, heatmap visualization, and AI-generated clinical insight make the system practically usable in healthcare. Experimental results show that the proposed system has reasonable predictive performance and modality-wise uniformity, overcoming significant limitations of traditional single-modality and black-box approaches.

However, some challenges remain: dependence on dataset quality, computational requirements for deep learning models, and the need for large and diverse datasets to improve generalization across different populations. Real-time deployment in resource-constrained environments may require additional optimization.

Future work could include multimodal fusion techniques for better integration of clinical and imaging data, transformer-based architectures and lightweight models to reduce computational complexity, and enhancement with longitudinal patient data, real-time monitoring, and integration into electronic health record (EHR) systems. Federated learning approaches could also be considered to maintain data privacy while allowing for joint model improvement. In this sense, NeuroXAI could become a robust and scalable clinical decision support system contributing to early diagnosis and management of Parkinson's Disease. Explainability combined with multimodal learning not only increases diagnostic performance but also bridges the gap between AI systems and clinical adoption through transparent and interpretable predictions.

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